

FinTech Transaction Analytics and Fraud Detection using SQL

A portfolio project demonstrating real-world financial data analysis, fraud detection, and business intelligence using SQL.





Project Overview

What It Analyzes

Financial transaction data to extract meaningful business insights — covering customer IDs, merchant categories, payment methods, amounts, status, and fraud indicators.

Why It Matters

Demonstrates real-world FinTech data analysis tasks using practical SQL concepts that mirror challenges faced by financial data teams daily.

What It Produces

Actionable business intelligence — from spending patterns and payment trends to fraud detection and revenue performance reporting.

Key Objectives

01

Build & Manage Databases

Create and manage SQL databases populated with realistic transaction data for analysis.

03

Detect Fraud

Identify fraudulent transactions and evaluate payment method performance using targeted queries.

02

Analyze Spending Patterns

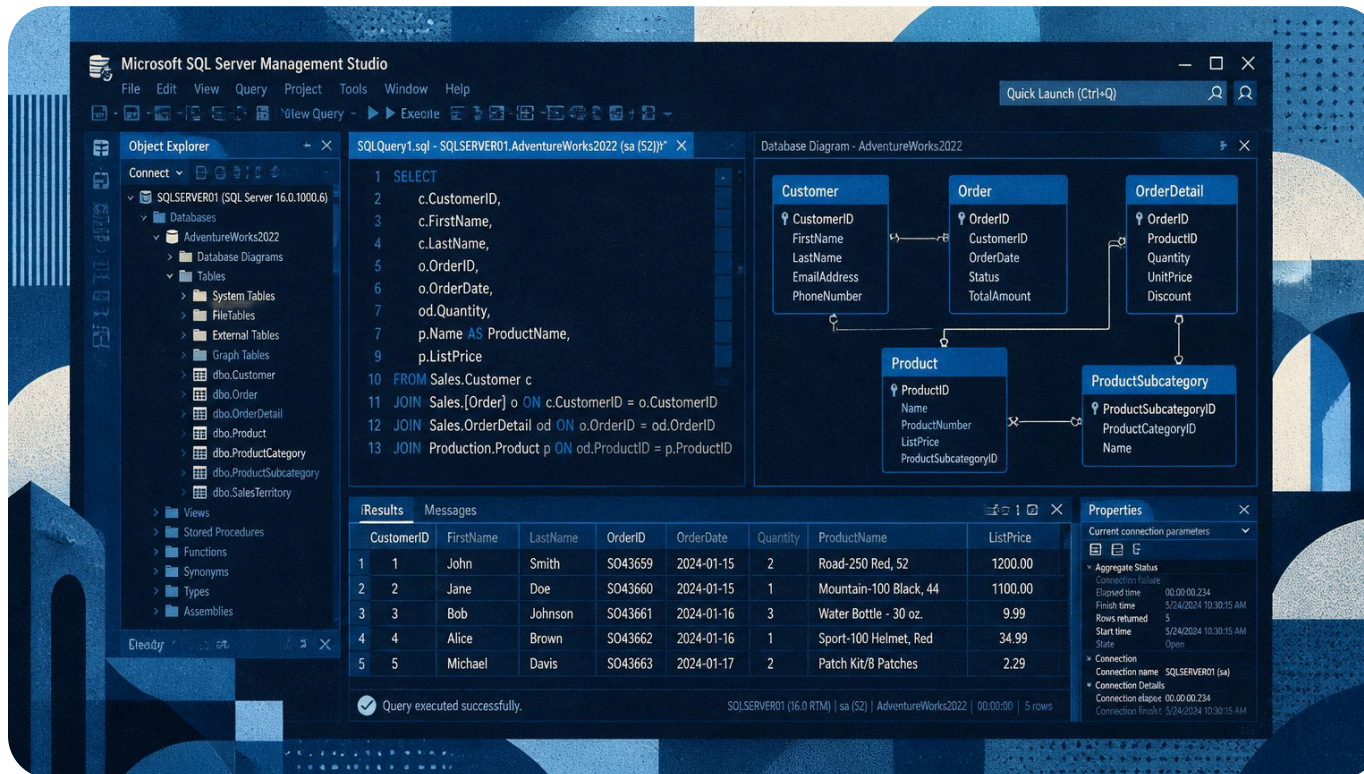
Examine customer spending behaviors and payment method trends across transaction records.

04

Generate Business Insights

Deliver actionable findings through well-crafted SQL queries that support financial decision-making.

SQL Concepts & Tools



TECHNOLOGIES

Microsoft SQL Server · Visual Studio Code · SQL

CORE SQL SKILLS

DDL (CREATE DATABASE, CREATE TABLE),
SELECT, WHERE, ORDER BY, GROUP BY

ADVANCED TECHNIQUES

Aggregate functions (COUNT, SUM, AVG, MAX, MIN), Window functions (RANK),
advanced filtering and multi-level
sorting

Business Problems Solved



Revenue Metrics

Calculate total transactions, revenue, and average transaction amounts across the dataset.



Top Customers

Identify highest and lowest value transactions and surface the top-spending customers by history.



Payment Analysis

Analyze payment method preferences and evaluate merchant category performance by revenue.



Fraud Detection

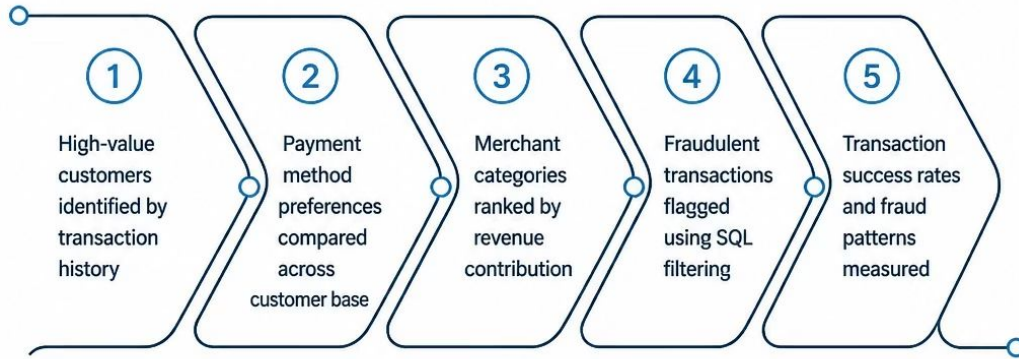
Detect fraudulent transactions and measure fraud percentage across the full dataset.



Success Rates

Evaluate transaction success and failure rates to support operational decision-making.

Key Insights Generated



From Raw Data to Decisions

Each insight was produced entirely through structured SQL queries — no external BI tools required. The project shows how a well-designed query pipeline can replace entire reporting workflows.

- Customer segmentation by transaction value
- Payment channel benchmarking
- Revenue attribution by merchant category
- Rule-based fraud flagging and rate measurement

Learning Outcomes & Impact

Practical SQL Mastery

Strengthens database management, data aggregation, filtering, and reporting skills through hands-on application.

FinTech Relevance

Demonstrates real-world financial analytics capabilities directly applicable to banking, payments, and risk teams.

Portfolio Ready

Ideal for SQL portfolio development and technical interview preparation — shows depth and business context.

Risk & Decision Support

Applicable to financial decision-making, fraud risk management, and operational performance monitoring.

